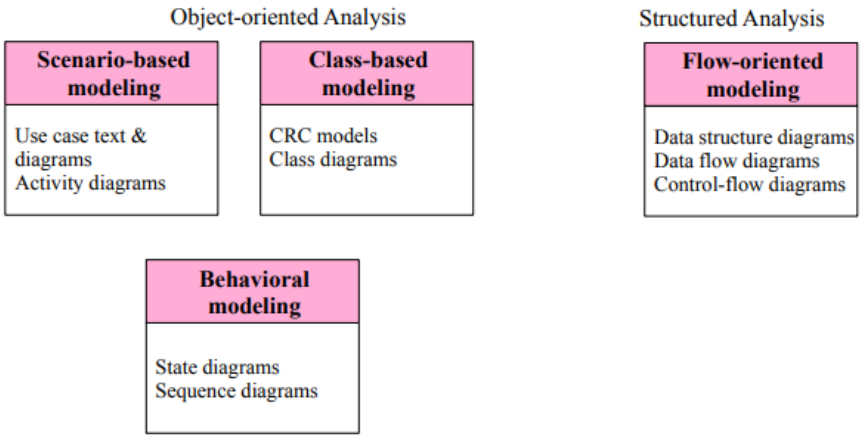
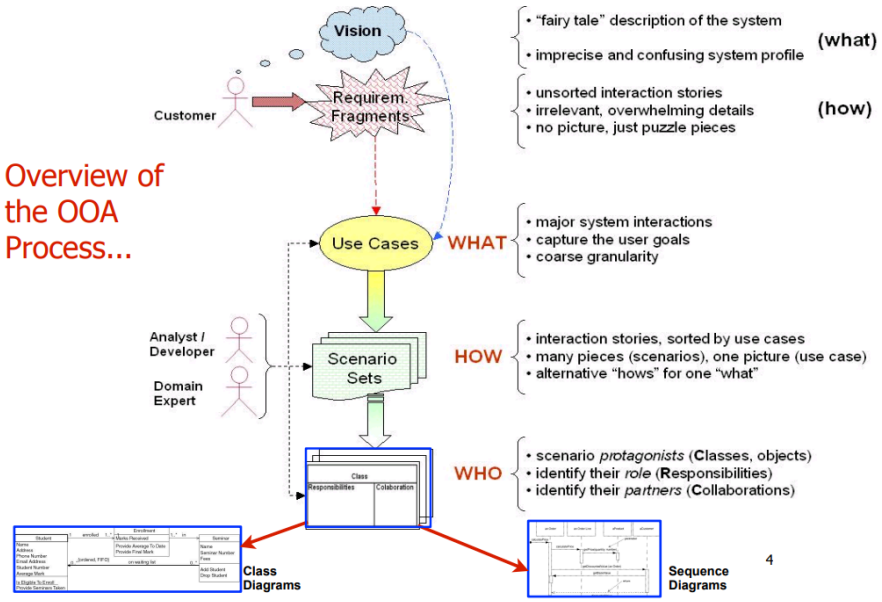
FIS curs 4 - Object-Oriented Analysis and Modeling

1.Object-Oriented Analysis and Modeling

- What are the relevant objects? How do they relate to one another?
- How do we specify/model a problem so that we can create an effective design?
- OOA aims to model the problem domain, by developing an object-oriented (OO) system.

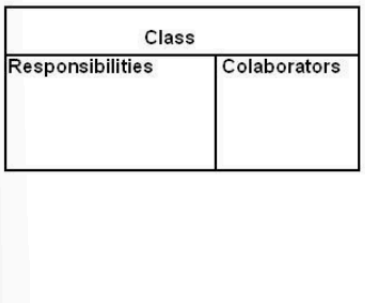
Elements of the Analysis Model

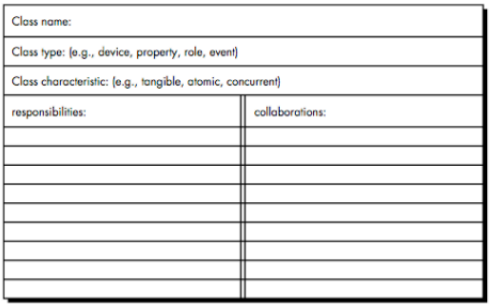




Class-Responsibility-Collaborator (CRC) Modeling

- Technique to identify candidate classes and indicate their responsibilities and collaborators
- Uses simple index cards





CRC Cards Session Scenario

- Use-Case driven
  - a use-case is the token
- Goal: be able to go through the whole use-case description by using the responsibilities written on the CRC cards
  - ...and of course following the Collaborator links

Rules for Identifying Classes

1.Retained Information

• information about object must be remembered for the system to function

2. Needed Services

• have a set of operations that change the value of its attributes

3. Multiple Attributes

• focus on “major” information
• object with single attribute is ok during design, but during analysis is just an attribute of another object

4. Essential Requirements objects

• entities that produce or consume information of the system, in any solution

Identifying Responsibilities

Attributes and operations of an identified class
Guidelines of Wirfs-Brock:

1. System intelligence should be evenly distributed

Operations = Verbs

- Computation
- Manipulation of data
  - e.g., add, delete, modify attributes
- Query

2. Information about one thing should be localized within a single class
3. Information and its related behavior should stay in the same class
4. Responsibilities should be shared among related classes

- about the state of an object
- Monitor an object
  - for the occurrence of a controlling event
- Has knowledge about the state of class and its associations

Collaborations

- Class can fulfill responsibilities by:
  1. using its own operations to manipulate its own attributes
  2. collaborating with others
- Three types of generic relationships:
  1. has-knowledge-of (association)
  2. is-part-of (aggregation)
  3. composition

Heuristics for Object-Oriented Modeling

Problema Proliferarii Claselor [Riel96]

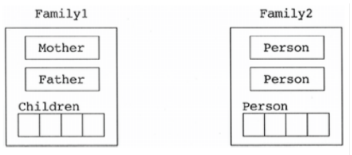
- Spaghetti Code vs. Ravioli Code
- Cum se manifesta "Codul Ravioli"
  - Vreau sa adaug o facilitate noua in sistem. Care 23 de clase din cele 4.200 de clase trebuie sa le modific?

Fiti retinuti in a modela ca si clase entitati din afara sistemului de implementat!

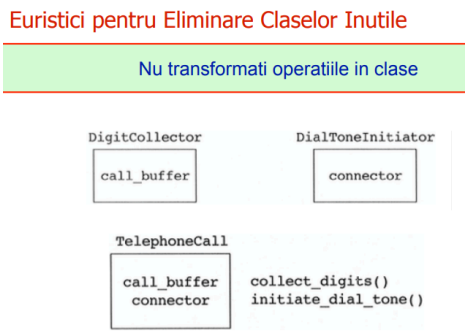
! Exemplu: Clientul unui Bancomat
! trimite un mesaj bancomatului

Clase sunt acelea care PRIMESC MESAJE (sunt apelate)
nu cele care TRANSMIT MESAJE (apeleaza)!

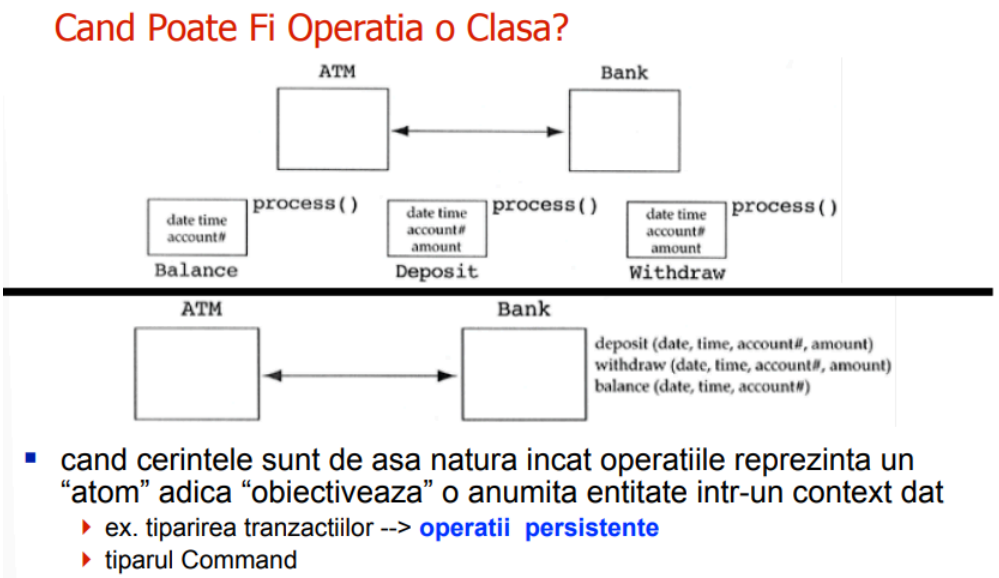
Roluri vs. Clase



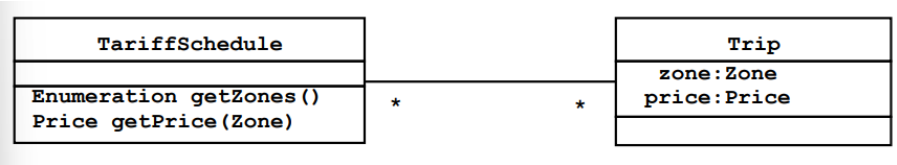
- Comportamentul este cel care decide!
  - daca comportamentul difera: avem clase;
  - daca nu: doar roluri ale aceleeasi clase
  - depinde de domeniul modelat de aplicatie



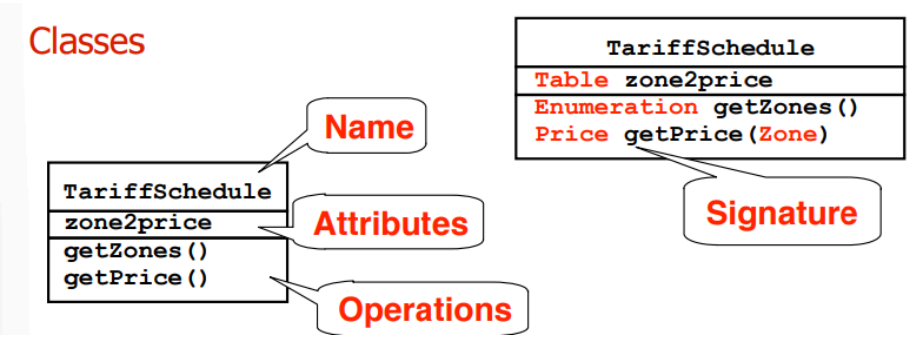
- naste() vs. schimbaScutece()



CLASS DIAGRAMS



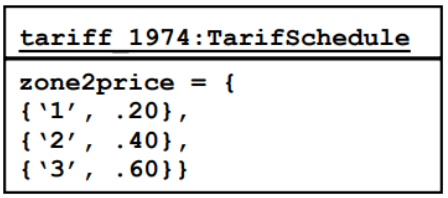
- Class diagrams represent the structure of the system.
- Class diagrams are used
  - during requirements analysis to model problem domain concepts
  - during system design to model subsystems and interfaces
  - during object design to model classes.



- A class represents a concept.
- contians state (attributes) and behavior (operations).

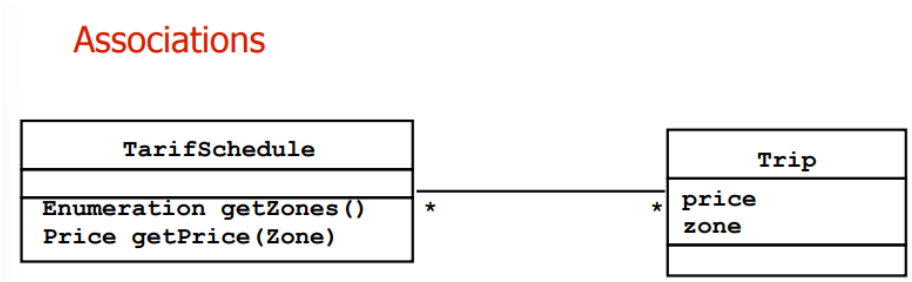
- Each attribute has a type.
- Each operation has a signature.
- The class name is the only mandatory information.

Instances



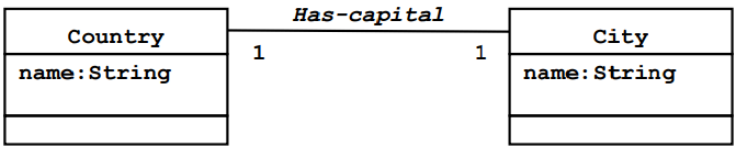
- An instance represents a phenomenon.
- The name of an instance is underlined and can contain the class of the instance.
- The attributes are represented with their values.

Associations

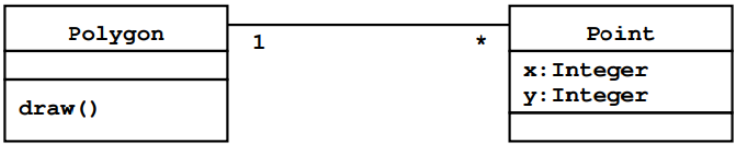


- Associations denote relationships between classes.
- The multiplicity of an association end denotes how many objects the source object can legitimately reference.

1-to-1 and 1-to-Many Associations

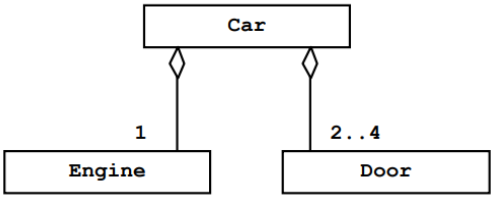


1-to-1 association



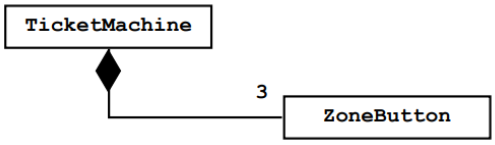
1-to-many association

Aggregation



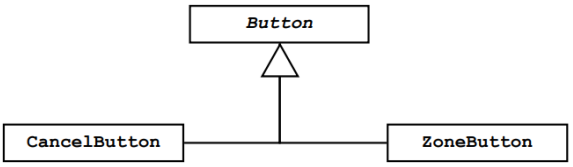
An aggregation is a special case of association denoting a
“consists of” (HAS-A) hierarchy.
The aggregate is the parent class, the components are the
children class.

Composition



A solid diamond denote composition, a strong form of
aggregation where components cannot exist without the
aggregate

Generalization



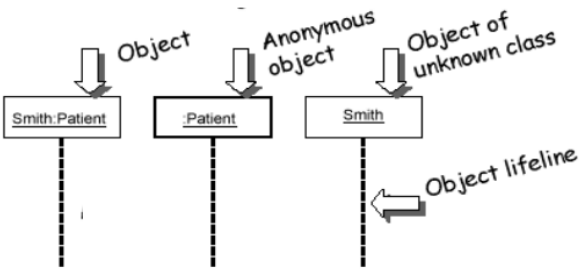
Generalization relationships denote inheritance between classes.
The children classes inherit the attributes and operations of the
parent class.
Generalization simplifies the model by eliminating redundancy.

Sequence Diagrams

UML Sequence Diagrams

- Sequence Diagram: an "interaction diagram" that models a single scenario executing in the system
  - perhaps 2nd most used UML diagram (behind class diagram)
- Participant: an object or entity that acts in the sequence diagram
  - sequence diagram starts with an unattached "found message" arrow
- Message: communication between participant objects
- Axes in a sequence diagram:
  - horizontal: which object/participant is acting
  - vertical: time (down → forward in time)

Representing Objects



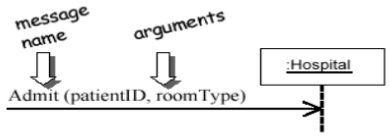
Name syntax: <objectname>:<classname>

- squares with object type, optionally preceded by object name and colon
  - write object's name if it clarifies the diagram
  - object's "life line" represented by dashed vert. line

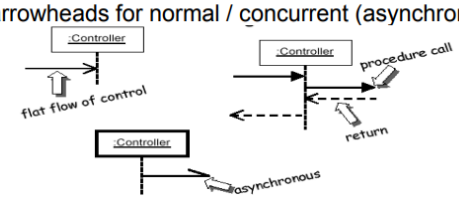
Messages between objects

message (method call) indicated by horizontal arrow to other object

- write message name and arguments above arrow

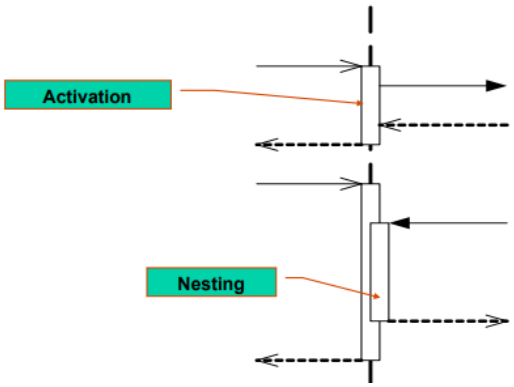


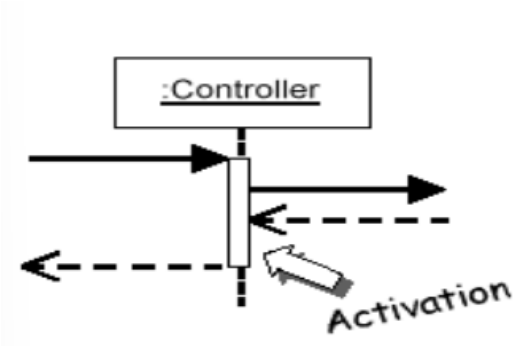
- dashed arrow back indicates return
- different arrowheads for normal / concurrent (asynchronous) methods



Indicating method calls

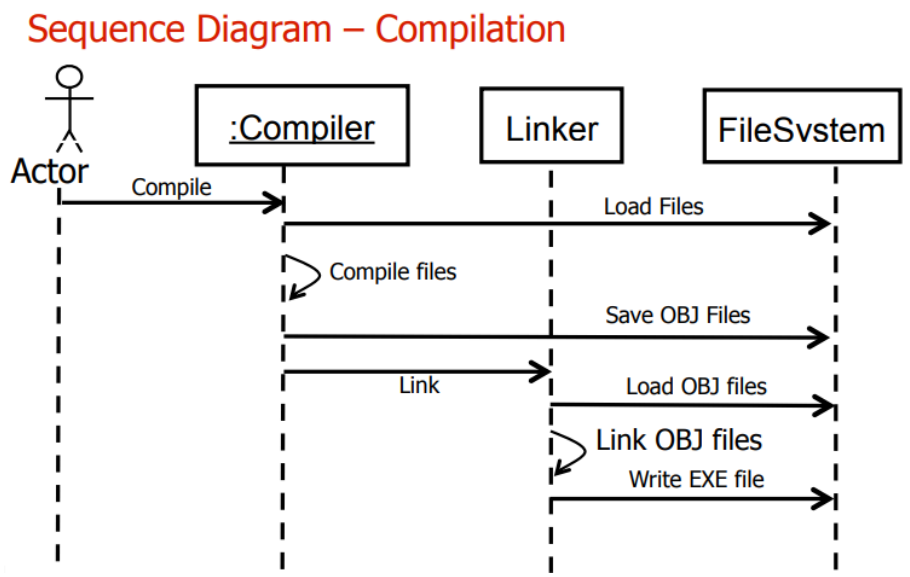
Activation: shows when object's method is on the stack
- either that object is running its code, or it is on the stack waiting for another object's method to finish
- nest to indicate recursion

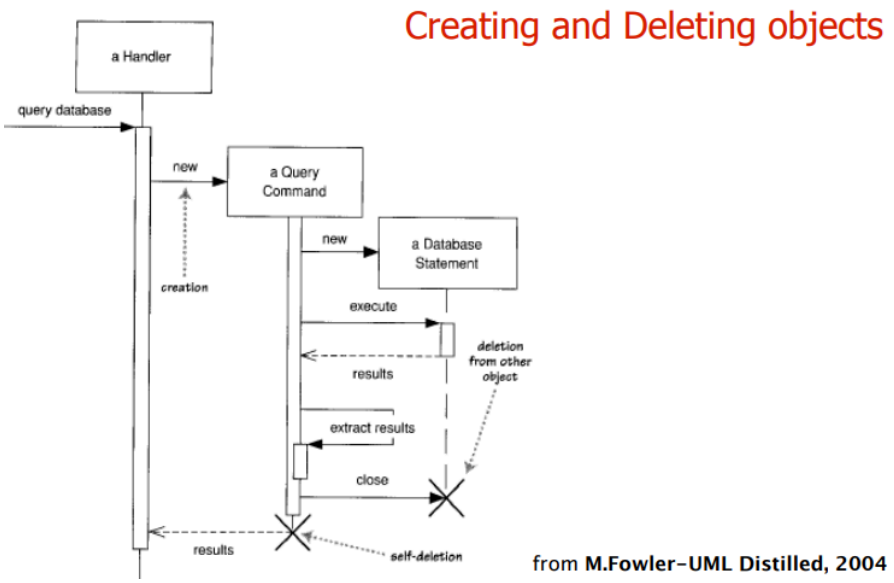




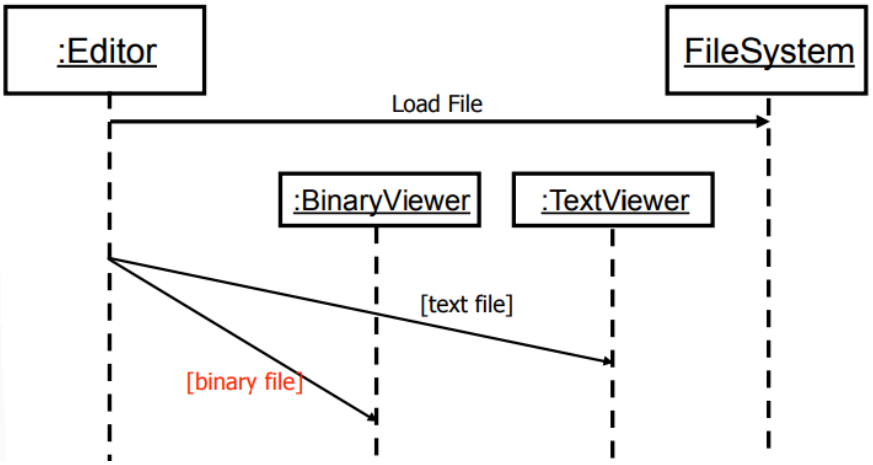
Example

- Building an executable from sources
  - load source files and compile them
  - load resulting object files and link them
  - write executable file

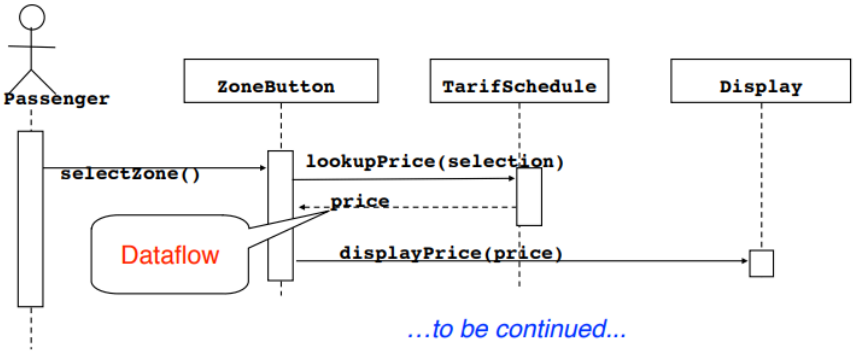




Branching Flow: flow goes to different objects [if condition is met]



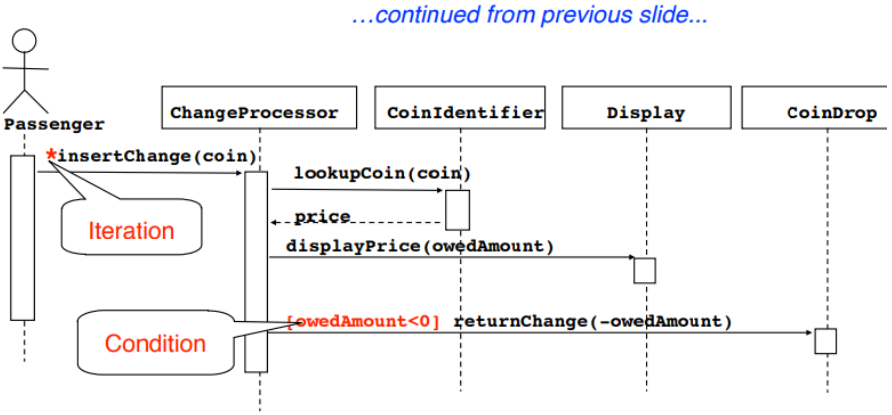
Flow of messages



- The source of an arrow indicates the activation which sent the message
- An activation is as long as all nested activations
- Horizontal dashed arrows indicate data flow
- Vertical dashed lines indicate lifelines

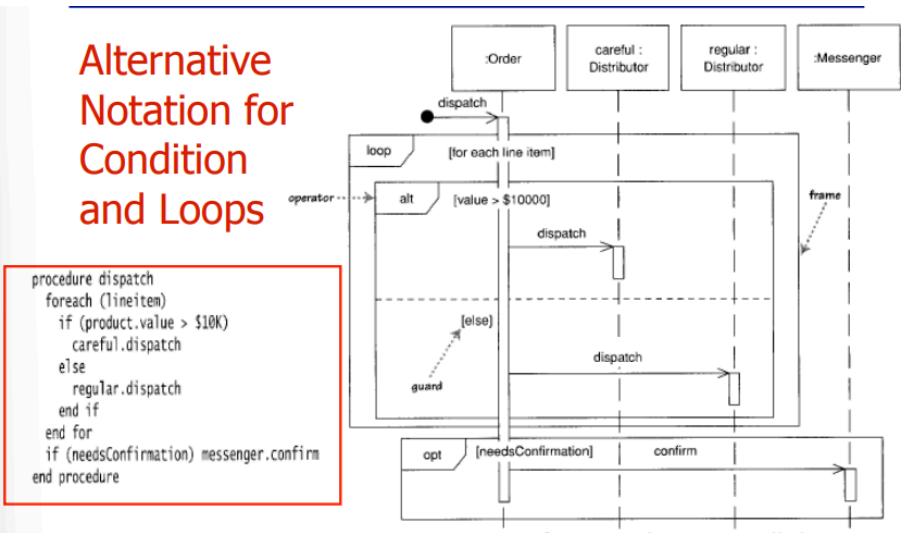
Iteration & condition

...continued from previous slide...



- Iteration is denoted by a * preceding the message name
- Condition is denoted by boolean expression in [ ] before the message name

Alternative Notation for Condition and Loops




```
procedure dispatch
  foreach (lineitem)
    if (product.value > $10K)
      careful.dispatch
    else
      regular.dispatch
    end if
  end for
  if (needsConfirmation) messenger.confirm
end procedure
```

from **M.Fowler–UML Distilled, 2004**

Sequence Diagram Summary

- UML sequence diagram represent behavior in terms of interactions.
- Useful to find missing objects.
- Time consuming to build but worth the investment.
- Complement the class diagrams (which represent structure).